

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and fabricated **biodegradable material formulations** from natural sources for porous electrodes, achieving a **15% reduction** in material waste through optimized laser writing processes.
- Developed **universal fabrication methodologies** for Laser Induced Graphene (LIG) on biopolymers, enabling electrode materials suitable for electrochemical sensing applications with **98% yield**.
- Integrated fabricated electrodes with green sensing active layers, facilitating the detection of biomarkers in textiles and contributing to a **30% improvement** in sensor sensitivity.
- Collaborated with cross-departmental teams and e-FLOWER project partners to develop novel integrated sensing e-textile platforms, reducing the environmental footprint by **20%**.
- Managed production support and incident response for enterprise Java/.NET applications, maintaining **99.95% SLA compliance** through proactive monitoring with **Splunk** and **Dynatrace**.
- Automated data validation and reporting tasks using **Python scripting** and **SQL queries** across PostgreSQL and Oracle databases, reducing manual effort by **40%**.

SKILLS

Materials Science & Fabrication – Laser Induced Graphene (LIG), Biodegradable Materials, Porous Electrodes, Nanomaterial Fabrication, Direct Laser Writing (DLW), Electrochemical Sensors, Supercapacitors.

Programming & Scripting – Python, Java, Spring Boot, SQL, R, Bash Scripting.

Data Analysis & BI – Power BI, DAX, Tableau, Power Query, Data Modelling, Statistical Analysis, NumPy, pandas, scikit-learn.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git.

Systems & Operations – Linux, Enterprise Systems Support, ERP Integration, Production Monitoring, Incident Response, Change Management, GDPR Awareness.

Research & Development – Materials Characterisation, Device Fabrication, Biomarker Detection, e-Textiles, Circular Economy Principles, Project Management.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Development of LIG-e-Textile Sensors for Environmental Monitoring

- Engineered novel **biodegradable material formulations** from sustainable sources, achieving a **25% increase** in porosity for enhanced LIG electrode performance.

- Implemented **Direct Laser Writing (DLW)** processes to fabricate scalable LIG electrodes on flexible biopolymers, meeting stringent **99% dimensional accuracy** requirements.
- Integrated fabricated electrodes with active sensing layers for real-time detection of environmental pollutants, demonstrating a **40% improvement** in detection limits compared to prior methods.

Automated Financial Transaction Monitoring Dashboard

- Developed a **payments operations dashboard** using Java Spring Boot and REST APIs, enhancing transaction-flow monitoring capabilities for a financial services client.
- Implemented **SQL queries** and data validation scripts in Python to support reconciliation workflows and ensure data integrity across PostgreSQL and Oracle databases.
- Configured **Splunk dashboards** for real-time operational alerting on failed or stalled payment messages, contributing to a **15% reduction** in manual oversight.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)